THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** -6	$a_n = a + (n-1)d$
2	**(d)** 84	Reverse AP has difference $-d$
3	**(a)** 3	$a_{n+1} - a_n = 3$
4	**(c)** 3	$a_1 = S_1$
5	**(b)** 36	$S_n = n^2$
6	**(c)** 5	$a + 9d$
7	**(a)** 78	$a_n = a + (n-1)d$
8	**(b)** 59	Reverse AP has difference $-d$
9	**(d)** 3	$a_{n+1} - a_n = 3$
10	**(a)** 4	$a_1 = S_1$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a_n = 3 + (n-1)(5)$ (1); $n = 13$ (1).	1+1
12	AP: $3, 6, 9, \ldots$ (1); $S_n = \frac{10}{2}[6 + 27] = 165$ (1).	1+1
13	$n = 12$ (1); $S = \frac{12}{2}(-5 + 28) = 138$ (1).	1+1
14	$a_n = 8 + (n-1)(3)$ (1); $n = 22$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a + d = 7$, $a + 4d = 19$ (2); subtracting, $d = 4$, $a = 3$ (2); $a_{10} = 39$ (1).	2+2+1
16	$a = 18$, $d = -\frac{5}{2}$, $l = -\frac{19}{2}$ (1); $l = a + (n-1)d$ gives $n - 1 = 11$ (1); $n = 12$ (1).	1+1+1
17	$a = 15$, $d = 4$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{22} = \frac{22}{2}[30 + 21(4)] = 1254$ (1).	1+1+1
	OR: $a + 2d = 11$, $a + 6d = 23$ (2); subtracting, $d = 3$, $a = 5$ (2); $a_{12} = 38$ (1).	2+2+1
18	$a = 6$, $d = 4$, $l = 86$ (1); $l = a + (n-1)d$ gives $n - 1 = 20$ (1); $n = 21$ (1).	1+1+1
	OR: $a = -9$, $d = 2$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{25} = \frac{25}{2}[-18 + 24(2)] = 375$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$S_n = \frac{n}{2}[2(24) + (n-1)(-2)] = 144$ (1); form quadratic in n (2); $n = 9$ (the other root is rejected) (2).	1+2+2
	OR: AP: $a = 100$, $d = 10$, $n = 20$ (1); last $= a + (n-1)d = 290$ (2); total $= S_n = \frac{20}{2}(100 + 290) = 3900$ (2).	1+2+2
20	$S_n = \frac{n}{2}[2(9) + (n-1)(8)] = 450$ (1); form quadratic in n (2); $n = 10$ (the other root is rejected) (2).	1+2+2
	OR: AP: $a = 200$, $d = 15$, $n = 12$ (1); last $= a + (n-1)d = 365$ (2); total $= S_n = \frac{12}{2}(200 + 365) = 3390$ (2).	1+2+2

Mistakes to avoid:

- Using $a_n = a + nd$ instead of $a + (n-1)d$.
- Writing $S_n = \frac{n}{2}[a + (n-1)d]$; the bracket needs $2a$.
- Taking d as $a_1 - a_2$; it is $a_2 - a_1$ (a decreasing AP has negative d).
- Using S_n when only a_n is asked (or the other way round) in word problems.
- Not rejecting a negative or fractional value of n obtained from a quadratic in n.
- Calling a list an AP after checking only the first two differences.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 64	$S_n = n^2$
2	**(d)** -20	$a + 9d$
3	**(a)** 17	$a_n = a + (n-1)d$
4	**(d)** 125	Reverse AP has difference $-d$
5	**(c)** 3	$a_{n+1} - a_n = 3$
6	**(c)** 5	$a_1 = S_1$
7	**(b)** 100	$S_n = n^2$
8	**(a)** 41	$a + 9d$
9	**(d)** 38	$a_n = a + (n-1)d$
10	**(c)** 72	Reverse AP has difference $-d$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	AP: $5, 10, 15, \ldots$ (1); $S_n = \frac{12}{2}[10 + 55] = 390$ (1).	1+1
12	$n = 15$ (1); $S = \frac{15}{2}(8 + 64) = 540$ (1).	1+1
13	$a_n = -2 + (n-1)(4)$ (1); $n = 10$ (1).	1+1
14	AP: $4, 8, 12, \ldots$ (1); $S_n = \frac{15}{2}[8 + 56] = 480$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a + 3d = 16$, $a + 5d = 28$ (2); subtracting, $d = 6$, $a = -2$ (2); $a_9 = 46$ (1).	2+2+1
16	$a = 9$, $d = 11$, $l = 174$ (1); $l = a + (n-1)d$ gives $n - 1 = 15$ (1); $n = 16$ (1).	1+1+1
17	$a = 8$, $d = \frac{1}{2}$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{20} = \frac{20}{2}[16 + 19(\frac{1}{2})] = 255$ (1).	1+1+1
	OR: $a + 4d = 2$, $a + 8d = -6$ (2); subtracting, $d = -2$, $a = 10$ (2); $a_{15} = -18$ (1).	2+2+1
18	$a = -7$, $d = 3$, $l = 80$ (1); $l = a + (n-1)d$ gives $n - 1 = 29$ (1); $n = 30$ (1).	1+1+1
	OR: $a = 33$, $d = -3$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{14} = \frac{14}{2}[66 + 13(-3)] = 189$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$S_n = \frac{n}{2}[2(30) + (n-1)(-3)] = 156$ (1); form quadratic in n (2); $n = 8$ (the other root is rejected) (2).	1+2+2
	OR: AP: $a = 500$, $d = 25$, $n = 10$ (1); last $= a + (n-1)d = 725$ (2); total $= S_n = \frac{10}{2}(500 + 725) = 6125$ (2).	1+2+2
20	$S_n = \frac{n}{2}[2(5) + (n-1)(4)] = 324$ (1); form quadratic in n (2); $n = 12$ (the other root is rejected) (2).	1+2+2
	OR: AP: $a = 50$, $d = 5$, $n = 16$ (1); last $= a + (n-1)d = 125$ (2); total $= S_n = \frac{16}{2}(50 + 125) = 1400$ (2).	1+2+2

Mistakes to avoid:

- Using $a_n = a + nd$ instead of $a + (n-1)d$.
- Writing $S_n = \frac{n}{2}[a + (n-1)d]$; the bracket needs $2a$.
- Taking d as $a_1 - a_2$; it is $a_2 - a_1$ (a decreasing AP has negative d).
- Using S_n when only a_n is asked (or the other way round) in word problems.
- Not rejecting a negative or fractional value of n obtained from a quadratic in n.
- Calling a list an AP after checking only the first two differences.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 3	$a_{n+1} - a_n = 3$
2	**(a)** 6	$a_1 = S_1$
3	**(d)** 144	$S_n = n^2$
4	**(b)** $\frac{31}{2}$	$a + 9d$
5	**(c)** -13	$a_n = a + (n-1)d$
6	**(a)** 42	Reverse AP has difference $-d$
7	**(c)** 3	$a_{n+1} - a_n = 3$
8	**(c)** 7	$a_1 = S_1$
9	**(a)** 225	$S_n = n^2$
10	**(b)** 27	$a + 9d$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$n = 10$ (1); $S = \frac{10}{2}(12 + -6) = 30$ (1).	1+1
12	$a_n = 20 + (n-1)(-3)$ (1); $n = 8$ (1).	1+1
13	AP: $7, 14, 21, \ldots$ (1); $S_n = \frac{20}{2}[14 + 133] = 1470$ (1).	1+1
14	$n = 20$ (1); $S = \frac{20}{2}(2 + 97) = 990$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a + 2d = 11$, $a + 7d = 36$ (2); subtracting, $d = 5$, $a = 1$ (2); $a_{11} = 51$ (1).	2+2+1
16	$a = 4$, $d = 6$, $l = 106$ (1); $l = a + (n-1)d$ gives $n - 1 = 17$ (1); $n = 18$ (1).	1+1+1
17	$a = 5$, $d = 8$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{18} = \frac{18}{2}[10 + 17(8)] = 1314$ (1).	1+1+1
	OR: $a + d = 9$, $a + 9d = 25$ (2); subtracting, $d = 2$, $a = 7$ (2); $a_{20} = 45$ (1).	2+2+1
18	$a = 12$, $d = 5$, $l = 117$ (1); $l = a + (n-1)d$ gives $n - 1 = 21$ (1); $n = 22$ (1).	1+1+1
	OR: $a = 21$, $d = 6$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{16} = \frac{16}{2}[42 + 15(6)] = 1056$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$S_n = \frac{n}{2}[2(17) + (n-1)(6)] = 517$ (1); form quadratic in n (2); $n = 11$ (the other root is rejected) (2).	1+2+2
	OR: AP: $a = 300$, $d = 20$, $n = 15$ (1); last $= a + (n-1)d = 580$ (2); total $= S_n = \frac{15}{2}(300 + 580) = 6600$ (2).	1+2+2
20	$S_n = \frac{n}{2}[2(45) + (n-1)(-4)] = 270$ (1); form quadratic in n (2); $n = 10$ (the other root is rejected) (2).	1+2+2
	OR: AP: $a = 1000$, $d = 50$, $n = 8$ (1); last $= a + (n-1)d = 1350$ (2); total $= S_n = \frac{8}{2}(1000 + 1350) = 9400$ (2).	1+2+2

Mistakes to avoid:

- Using $a_n = a + nd$ instead of $a + (n-1)d$.
- Writing $S_n = \frac{n}{2}[a + (n-1)d]$; the bracket needs $2a$.
- Taking d as $a_1 - a_2$; it is $a_2 - a_1$ (a decreasing AP has negative d).
- Using S_n when only a_n is asked (or the other way round) in word problems.
- Not rejecting a negative or fractional value of n obtained from a quadratic in n.
- Calling a list an AP after checking only the first two differences.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.